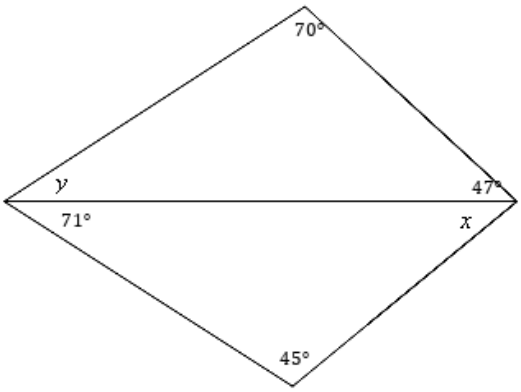
Angles of polygons

1 State the sum of the interior angles in a quadrilateral.

2 State the sum of exterior angles for any polygon.

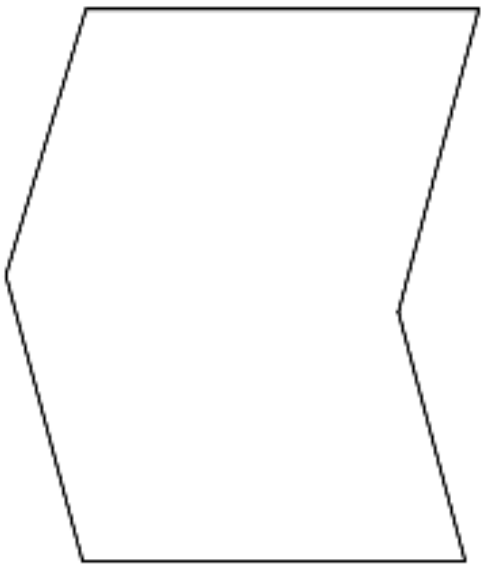
3 Consider the quadrilateral in the adjacent diagram.



a Find the value of the angle marked y.

b Find the value of the angle marked x.

4 Consider the non-convex hexagon in the diagram shown.

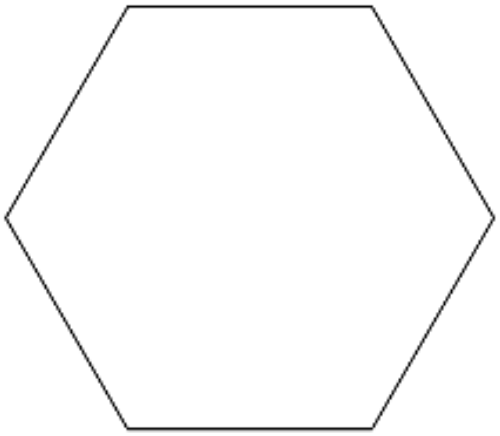


a Find the least number of triangles that it can be divided into.

b Determine the interior angle sum of the non-convex hexagon.

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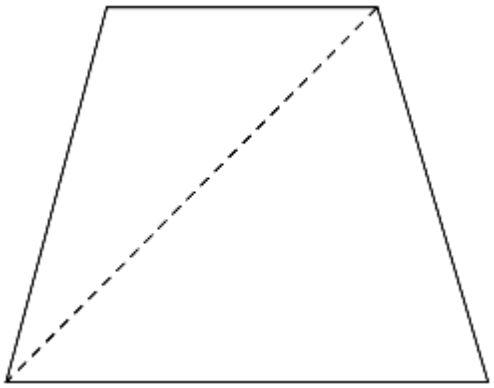
5 Consider the convex hexagon in the adjacent diagram.



a Find the least number of triangles that the hexagon can be divided into.

b Determine the interior angle sum of the hexagon.

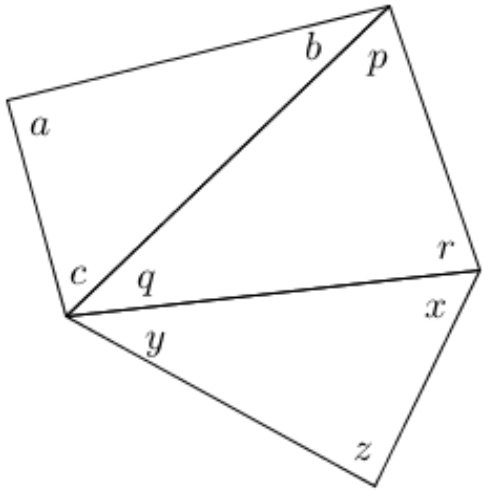
6 Consider the adjacent diagram.
The quadrilateral is divided into two triangles (this is the least number of triangles that a quadrilateral can be divided into). The interior angle sum of a quadrilateral is therefore equal to double the interior angle sum of a triangle.
Complete the following table.



Name of the polygon	Number of sides	Least number of triangles	Interior angle sum
Pentagon			
Hexagon			
Heptagon			
Octagon			
Nonagon			
Decagon			

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7 Consider the pentagon to the right.



a Find the value of the following:

i $a+b+c$

ii $p+q+r$

iii $x+y+z$

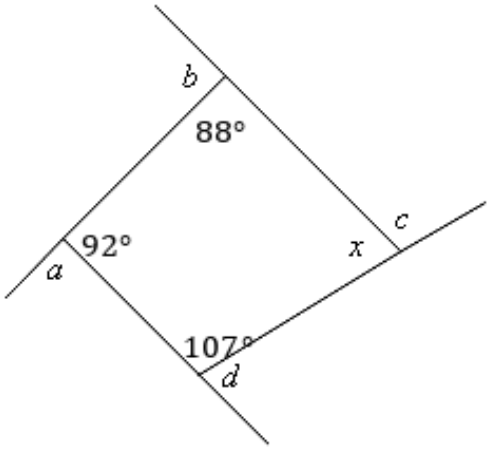
b Determine the angle sum of the pentagon.

c How many non-overlapping triangles can an n-sided figure be split into?

d Hence or otherwise, state the interior angle sum of an n-sided polygon.

8 Find the interior angle sum of a regular polygon whose exterior angles measure 40°.

9 Consider the adjacent quadrilateral.



a Find the value of the following angles:

i x **ii** a

iii b **iv** c

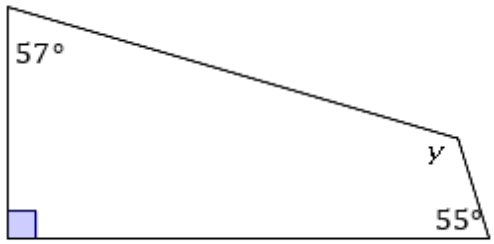
v d

b State the sum of exterior angles in a quadrilateral.

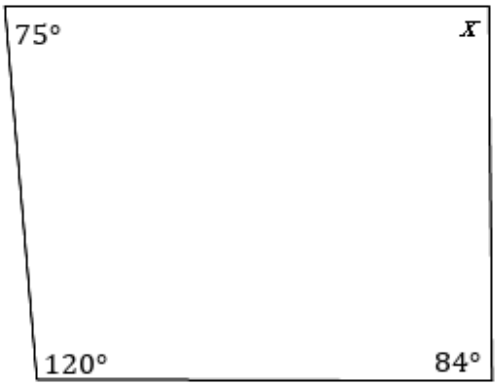
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10 Find the value of the variable(s) in the following diagrams.

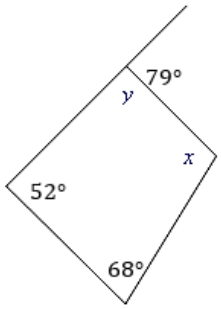
a



b

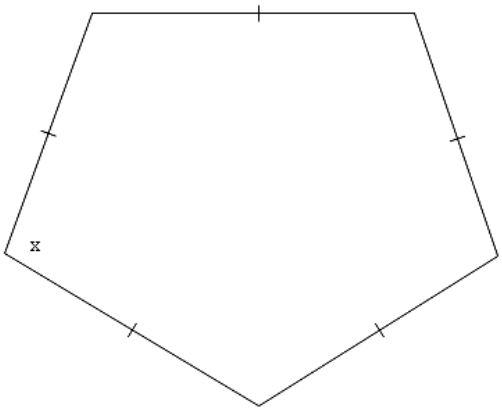


c

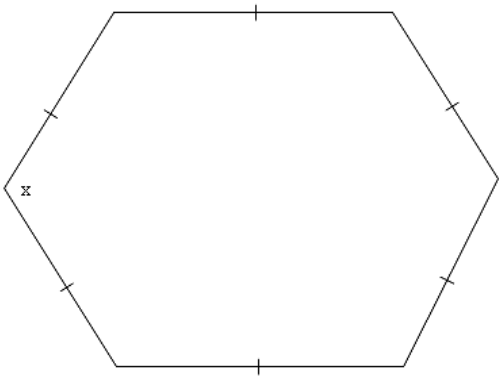


11 Given that each of the following polygon is regular, find the value of x.

a



b



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12 Consider the following diagram.
Calculate x.



13 The interior angles of an n-sided regular polygon each have a measure of x°.
Write down a formula for x in terms of n.

14 Find the measure of each of the interior angles in a regular:

a 6-sided polygon

b Octagon

c Decagon

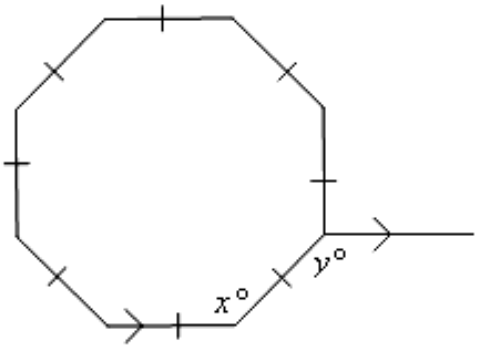
d Dodecagon

15 Find the measure of each exterior angle of a regular polygon with 20 sides.

16 Consider the regular polygon in the adjacent diagram.

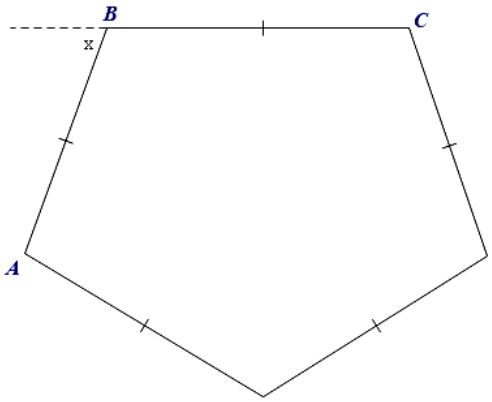
a Solve for x.

b Solve for y.



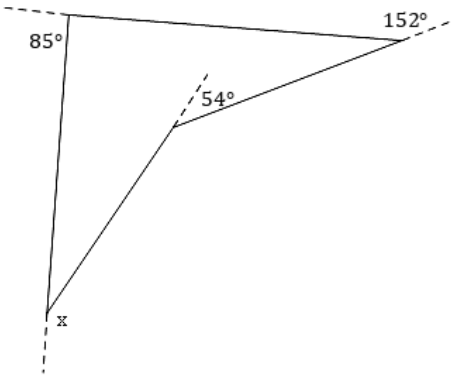
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17 Given that the shape in the diagram is a regular pentagon, find the value of x.

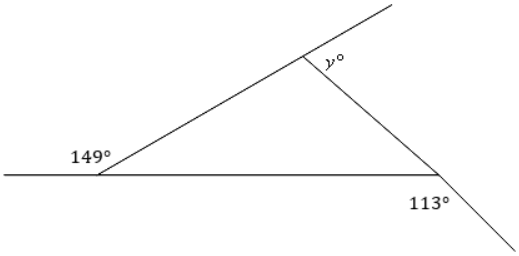


18 Find the value of the variable in the following diagrams.

a



b



19 The exterior angles of an n-sided regular polygon each have a measure of x°. Write down a formula for x.

20 Given the following angle measures of a regular polygon:

i Find the number of sides of the polygon.

ii Identify the shape of the polygon if it exists.

a Interior angles equal to 150° each.

b Interior angles equal to 140° each.

c Interior angles equal to 120° each.

d Exterior angles equal to 36° each.

e Exterior angles equal to 72° each.

f Exterior angles equal to 45° each.

g Exterior angles equal to 75° each.

h Exterior angles equal to 50° each.

i Interior angles equal to 100° each.

j Interior angles equal to 130° each.

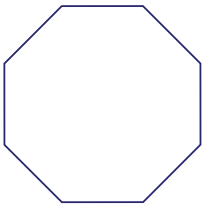
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Additional questions

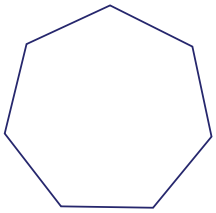
21 Find the measure of an interior angle in each of the following regular polygons:

a Hexagon

b



c

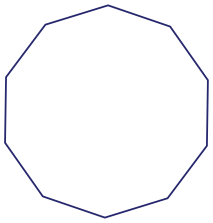


d 22-gon

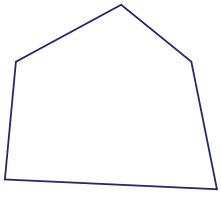
22 Find the sum of the interior angles in each of the following polygons:

a Decagon

b



c



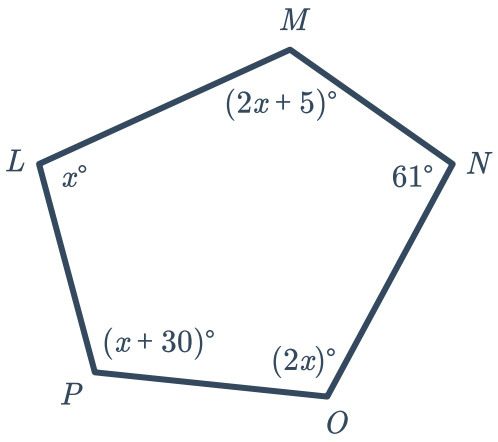
d 22-gon

23 What is the sum of the exterior angles of a 13-gon?

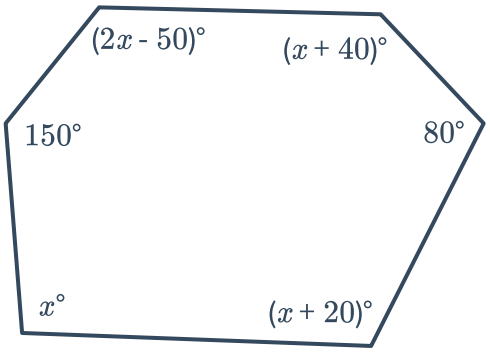
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24 Solve for x in the following polygons:

a

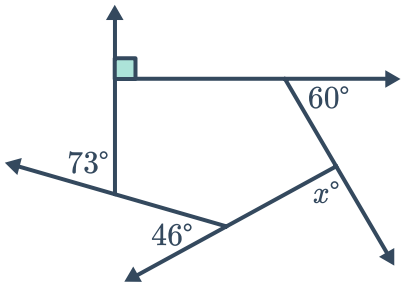


b

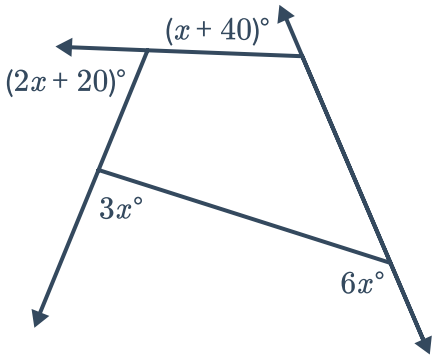


25

a Solve for x.

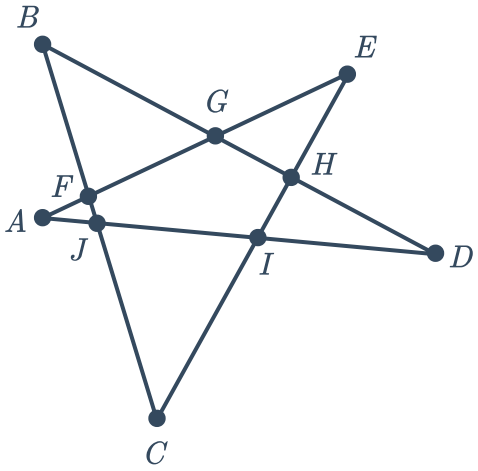


b



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26 In the diagram, $m\angle B = 43^\circ$, $m\angle FJI = 109^\circ$, $m\angle GHI = 92^\circ$, $m\angle JFG = 101^\circ$, and $m\angle C = 45^\circ$.



Solve for the following:

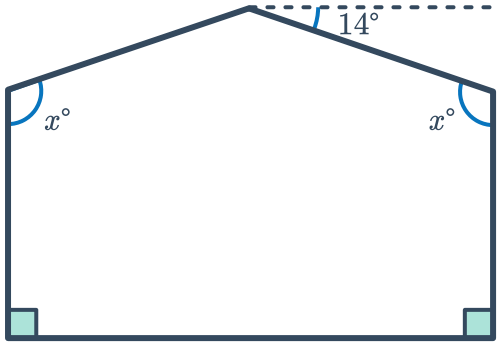
a $m\angle A$

b $m\angle FGH$

c $m\angle HIJ$

d $m\angle BFG + m\angle EGH + m\angle DHI + m\angle CIJ + m\angle AJF$

27 A house is built with a pitched roof. The engineer states that the angle of the pitch is 14°. Solve for x.

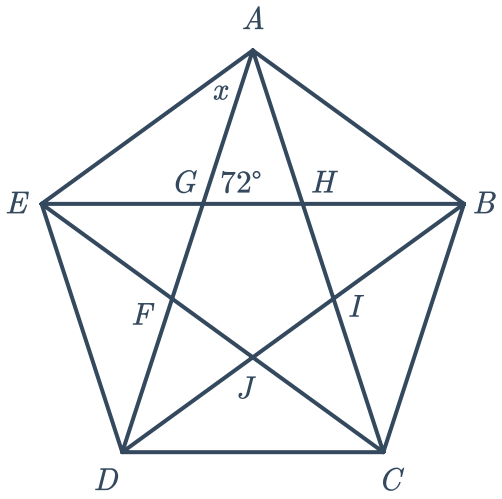


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28 Determine the number of sides of a polygon when:

- **a** The interior angles sum to 360°
- **b** Each interior angle has a measure of 135°
- **c** Each exterior angle has a measure of 24°
- **d** Each exterior angle has a measure of 15°
- **e** The interior angles sum to 1980°
- **f** Each interior angles has a measure of 160°

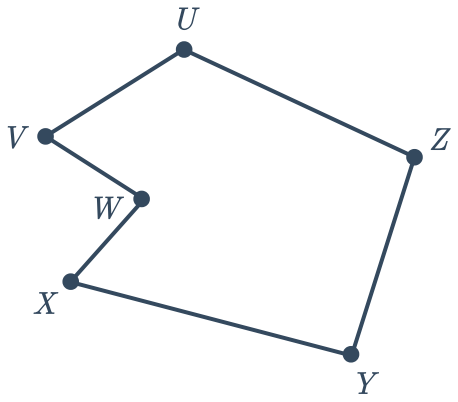
29 $ABCDE$ is a regular pentagon.



- **a** What is the measure of $\angle AED$? Explain how you know.
- **b** Explain why $x = 36^\circ$.

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30 Given: Hexagon $UVWXYZ$



a Divide the hexagon into six triangles, all of which share a common vertex T on the interior of $UVWXYZ$.

b Show that the sum of the angles in the six triangles is 1080°.

c Show that the sum of the six angles that meet at T is 360°.

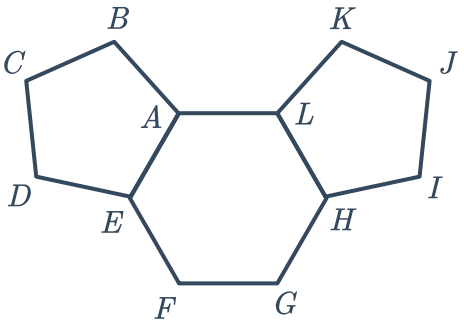
d Using the results from parts a) and b), show that the sum of the interior angles of Hexagon $UVWXYZ$ is 720°.

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31 The pattern on a soccer ball is a tesselation, or a repeated pattern with no overlaps or gaps. On a three-dimensional soccer ball, the tesselation consists of regular pentagons and hexagons:



When laid flat, a piece of the pattern looks like this:



Explain why the pattern on a soccer ball is impossible to tesselate on a two-dimensional surface.

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